

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent, suspension	Molecules Electroporated	DNA: linearized plasmids, about 10 kB.
Species Used	Hamster, CHO, ovary		

Before the Pulse

Cell Growth Medium	Hams F12 (Gibco)	Growth Phase at Harvest	70 % confluence
		Pre-pulse Incubation	10 min. on ice

Wash Solution Phosphate Buffered Saline, pH 7.4

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	0 to 4 °C		
Electroporation Medium*	Phosphate Buffered Saline (PBS), pH 7.4	Cuvette Gap	0.4 cm
Cell Density	5x10 (6) cells / 700 µl	Voltage	0.75 kV
Volume of Cells	700 µl	Field Strength	1.88 kV/cm
DNA Concentration	20 to 100 µg / 700 µl		
DNA Resuspension Buffer	PBS or TE (10 mM Tris, 1 mM EDTA, pH 8.0)	Capacitor	25 µF
Volume of DNA	20 µl to 50 µl; (conc.=1µg / µl)	Resistor	(Pulse Controller) Ω none
		Time Constant	0.5 msec

After the Pulse

Outgrowth Medium Hams F12 (Gibco)

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
PBS: 1x = 8g NaCl, 0.2g KCl, 0.2g KH₂PO₄, 1.15g Na₂HPO₄
HBS: 10mM HEPES, pH 7.2, 150 mM NaCl, 5 mM CaCl₂

Outgrowth Temperature	37 °C
Length of Incubation	2 days before selection
Selection Method or Assay Used	F12 - 5158 plus or minus MTX
Electroporation Efficiency	250 to 1000 transformants / µg DNA
Per Cent Survival	25%

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Survey Number

088